

Operations Research

Lecture 3: Solving Linear Programs

A textbook treatment

Based on the lecture notes of El Mehdi Saad

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How to read this book

Each concept in this book is presented in the same rhythm.

- 1. Motivation and intuition.** Why do we care? What problem does the idea solve, and what picture should you hold in your head?
- 2. Formal statement.** The precise definition or theorem, set in a coloured box so it is easy to find when revising.
- 3. Proof.** Every claim is proved. Longer proofs are split into labelled steps, and the one idea that makes the proof work is isolated in a *Key insight* box.
- 4. Interpretation and discussion.** What the result really says, where it applies, and which misreadings to avoid.
- 5. Examples.** From “trivially small” to “genuinely useful.”
- 6. Techniques, classic problems and gotchas.** Computational shortcuts, proof patterns, exam favourites with one-line solutions, and the mistakes everyone makes once.

Margin notes carry the informal, “say it out loud” version of what is happening in the main text. Every major section closes with a summary box.

Prerequisites. Comfort with vectors, matrices, systems of linear equations, row reduction and the notions of rank and linear independence. Chapter 2 recalls exactly what is needed as it goes.

Chapter 3

Solving Linear Programs: The Simplex Method

3.1 Preparing an LP: Redundancy and Two Routes to Equational Form

Motivation

Chapter 2 established the pipeline: model in canonical form, convert to equational form, and search among basic feasible solutions. Before the conversion there is an optional but useful step. Some inequalities in a model are *implied* by the others: they exclude nothing that was not already excluded. Removing them changes neither the feasible set nor the optimum, but it reduces the number of slack variables and therefore the number of column selections to examine. This section makes the test precise, then solves one LP both with and without the removal to see exactly what changes and what does not.

Every inequality you keep becomes a slack variable, a column, and more candidate bases. Dropping an inequality that says nothing new is free.

Recall the running example: a workshop chooses quantities x_1, x_2 of two products,

$$\max 2x_1 + x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 4, \quad x_1 \leq 3, \quad 2x_1 + x_2 \leq 7, \quad x_1, x_2 \geq 0.$$

Adding the first two inequalities gives $2x_1 + x_2 \leq 7$, which is exactly the third. The third inequality is therefore redundant: every point satisfying the first two automatically satisfies it.

Proposition 3.1.1 (Redundancy test). *Let $a_i^\top x \leq b_i$ ($i = 1, \dots, m$) be a system of inequalities and let $a^\top x \leq b$ be a candidate inequality. If there exist $\lambda_1, \dots, \lambda_m \geq 0$ with*

$$\sum_i \lambda_i a_i = a \quad \text{and} \quad \sum_i \lambda_i b_i \leq b,$$

then every x satisfying the system satisfies the candidate; the candidate is redundant and may be removed without changing the feasible set. Variable bounds such as $x_j \geq 0$ may be used as inequalities $-x_j \leq 0$ in the combination.

Proof. For any x satisfying the system, $a^\top x = \sum_i \lambda_i a_i^\top x \leq \sum_i \lambda_i b_i \leq b$, where the first inequality uses $\lambda_i \geq 0$ (multiplying an inequality by a nonnegative number preserves it) and the second is the hypothesis. So the candidate holds automatically. Removing it leaves the feasible set unchanged because the set of x satisfying “system and candidate” equals the set satisfying “system”. $\square$

Key insight

Redundancy is a *proof*: a nonnegative combination of the other inequalities that dominates the candidate. It is the same certificate mechanism as weak duality (Lemma 1.3.10), pointed at a constraint instead of at the objective.

Example 3.1.2 (Three variables, five inequalities). Which inequality can be removed from

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4, \quad x_2 + x_3 \leq 5, \quad x_1 + 2x_3 \leq 6, \\ & x_3 \leq 2, \quad 2x_1 + 3x_2 + x_3 \leq 14, \quad x \geq 0 ? \end{aligned}$$

Take $\lambda = (2, 1, 0, 0)$ on the first four: $2(x_1 + x_2) + (x_2 + x_3) = 2x_1 + 3x_2 + x_3$, and $2 \cdot 4 + 5 = 13 \leq 14$. The fifth inequality is redundant; the others imply the even tighter bound 13. Removing it leaves four inequalities and hence four slacks instead of five.

Technique: Spotting redundancy fast

- Try to *match the left-hand side* first: which combination of rows produces the candidate's coefficients? Only then compare right-hand sides.
- Small integer multipliers are the norm in exercises: $1+1$, $2 \cdot (\text{row}) + (\text{row})$.
- If the candidate has a coefficient that no other row can supply (e.g. x_3 appears only in the candidate), it cannot be redundant.
- Do not remove an inequality that is merely *loose at the optimum*; redundancy is about the whole feasible set, not about one point.

Route A: remove the redundant inequality, then add slacks

Example 3.1.3 (Route A). Keeping $x_1 + x_2 \leq 4$ and $x_1 \leq 3$ only, add slacks $s_1, s_2 \geq 0$:

$$x_1 + x_2 + s_1 = 4, \quad x_1 + s_2 = 3,$$

so that, with $z = (x_1, x_2, s_1, s_2)^\top$,

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

The slack columns form I_2 , so $\text{rank } A = 2$. There are $\binom{4}{2} = 6$ column selections. One in detail: select x_1, x_2 and set $s_1 = s_2 = 0$:

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & 0 & 3 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & -1 & -1 \end{array} \right],$$

so $x_2 = 1$, $x_1 = 3$, $z = (3, 1, 0, 0) \geq 0$, profit $2 \cdot 3 + 1 = 7$. Another: select x_2, s_1 .

Their columns are $(1, 0)^\top$ and $(1, 0)^\top$, so $M = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ has rank $1 < 2$: not a basis; skip it (the columns stay in the LP and may belong to other bases). All six:

Selected	(x_1, x_2, s_1, s_2)	Feasible?	Profit
x_1, x_2	$(3, 1, 0, 0)$	yes	7
x_1, s_1	$(3, 0, 1, 0)$	yes	6
x_1, s_2	$(4, 0, 0, -1)$	no	
x_2, s_1	dependent columns		
x_2, s_2	$(0, 4, 0, 3)$	yes	4
s_1, s_2	$(0, 0, 4, 3)$	yes	0

Six selections, five bases, four distinct BFSs. The best BFS is $(x_1, x_2) = (3, 1)$ with $v^* = 7$. Comparing BFSs suffices because the LP is feasible and bounded (Theorem 2.5.2); we enumerated all BFSs, not all feasible points.

Route B: keep every inequality

Example 3.1.4 (Route B). With one slack per inequality,

$$\tilde{A} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix}, \quad z = (x_1, x_2, s_1, s_2, s_3)^\top.$$

I_3 in the slack columns guarantees rank 3; nothing to delete. Now $\binom{5}{3} = 10$ selections. One in detail: select x_1, x_2, s_3 , set $s_1 = s_2 = 0$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 1 & 0 & 0 & 3 \\ 2 & 1 & 1 & 7 \end{array} \right] \xrightarrow[R_3 \leftarrow R_3 - R_1 - R_2]{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right],$$

so $x_2 = 1$, $x_1 = 3$, $s_3 = 0$. Three pivots, so the columns are independent; the BFS is $(3, 1, 0, 0, 0)$. Note that s_3 is basic yet equals zero: a *degenerate* BFS. All ten selections:

Selected	$(x_1, x_2, s_1, s_2, s_3)$	Feasible?	Profit
x_1, x_2, s_1	$(3, 1, 0, 0, 0)$	yes	7
x_1, x_2, s_2	$(3, 1, 0, 0, 0)$	yes	7
x_1, x_2, s_3	$(3, 1, 0, 0, 0)$	yes	7
x_1, s_1, s_2	$(\frac{7}{2}, 0, \frac{1}{2}, -\frac{1}{2}, 0)$	no	
x_1, s_1, s_3	$(3, 0, 1, 0, 1)$	yes	6
x_1, s_2, s_3	$(4, 0, 0, -1, -1)$	no	
x_2, s_1, s_2	$(0, 7, -3, 3, 0)$	no	
x_2, s_1, s_3	dependent columns		
x_2, s_2, s_3	$(0, 4, 0, 3, 3)$	yes	4
s_1, s_2, s_3	$(0, 0, 4, 3, 7)$	yes	0

Ten selections, nine bases, six feasible bases, four distinct BFSs. Three different bases produce the same degenerate vertex $(3, 1)$.

Proposition 3.1.5 (The two routes are equivalent). *In Examples 3.1.3 and 3.1.4, the map $(x_1, x_2, s_1, s_2) \mapsto (x_1, x_2, s_1, s_2, s_1 + s_2)$ is a bijection between the feasible sets of Route A and Route B that preserves the objective. Hence both routes have the same optimal decisions and value.*

Proof. For any feasible point of Route B, $s_3 = 7 - 2x_1 - x_2 = (4 - x_1 - x_2) + (3 - x_1) = s_1 + s_2$, so s_3 is determined by the other variables and is automatically ≥ 0 when $s_1, s_2 \geq 0$. Thus dropping s_3 gives a feasible point of Route A, and adding $s_3 = s_1 + s_2$ recovers it; the two operations are inverse. The objective involves only x_1, x_2 . $\square$

	Route A: remove	Route B: keep
Slack variables	2	3
Column selections	6	10
Bases	5	9
Feasible bases	4	6
Distinct BFSs	4	4

Interpretation & discussion

Same optimal decision $(3, 1)$, same profit 7, same four corners of the polygon. Route B paid for the redundant inequality with more variables and more selections, and it introduced degeneracy: the vertex $(3, 1)$ lies on three constraint lines ($x_1 + x_2 = 4$, $x_1 = 3$, $2x_1 + x_2 = 7$) in a plane where two suffice, so three bases describe it. Removal is optional, but here it reduces the work and removes the degeneracy.

Summary: Redundancy and the two routes

- An inequality that is a nonnegative combination of others (with a no-larger right-hand side) is redundant (Proposition 3.1.1).
- Recipe: optionally remove redundant inequalities; add one slack per retained inequality; $A = [G \ I_m]$ has rank m ; choose m independent columns, solve, check signs; compare BFSs.
- Keeping a redundant inequality changes nothing about the optimum but adds variables, selections and degeneracy.

3.2 Enumeration: Correct but Expensive

Motivation

The fundamental theorem (Theorem 2.5.2) turns an infinite search into a finite one. This section records what that finite search costs, checks the two hypotheses of the theorem on examples, and shows what happens when boundedness fails. The conclusion motivates the simplex method: the theorem narrows the search, but does not make enumeration efficient.

Theorem 3.2.1 (Fundamental theorem, restated). *Let $A \in \mathbb{R}^{m \times n}$ have full row rank and consider $\max c^\top z$ s.t. $Az = b$, $z \geq 0$. If the feasible set is nonempty and the objective is bounded above, there is an optimal basic feasible solution. (Proof: Theorem 2.5.2.)*

Technique: Enumeration algorithm

1. Select m columns of A . Skip dependent selections.
2. Set $z_N = 0$ and solve $A_B z_B = b$. Keep the solution only if $z_B \geq 0$.
3. Compare the objective values of all distinct BFSs; the best is optimal.

The number of candidates satisfies

$$\#\{\text{distinct BFSs}\} \leq \#\{\text{bases}\} \leq \binom{n}{m}.$$

Example 3.2.2 (The count explodes). $\binom{4}{2} = 6$, $\binom{5}{3} = 10$, $\binom{20}{10} = 184,756$. A model with 20 decisions and 20 inequalities has $n = 40$ variables after slacks and $\binom{40}{20} = 137,846,528,820$ column selections. Each requires solving a 20×20 system. Exhaustive enumeration is impractical at that scale, and industrial LPs have millions of variables.

Checking the hypotheses

Example 3.2.3 (A bounded objective: the running example). For $\max 2x_1 + x_2$ with $x_1 + x_2 \leq 4$, $x_1 \leq 3$, $x \geq 0$: feasibility holds since $(0,0)$ satisfies everything. Boundedness: adding the two resource inequalities, $2x_1 + x_2 = (x_1 + x_2) + x_1 \leq 4 + 3 = 7$, so $U = 7$ is a finite upper bound and the theorem applies. The feasible point $(3,1)$ attains 7, so it is optimal by matching bounds (Theorem 1.5.4).

Proposition 3.2.4 (A feasible improving ray proves unboundedness). *Let x^0 be feasible for $\max c^\top x$ over a feasible set X , and let d be a direction such that $x^0 + \theta d \in X$ for all $\theta \geq 0$ and $c^\top d > 0$. Then the LP is unbounded.*

Proof. $c^\top(x^0 + \theta d) = c^\top x^0 + \theta c^\top d \rightarrow +\infty$ as $\theta \rightarrow \infty$, and every point of the ray is feasible, so no finite M bounds the objective over X . $\square$

Example 3.2.5 (When boundedness fails). $\max x_1 + x_2$ s.t. $x_1 \leq 3$, $x \geq 0$. The ray $(x_1, x_2) = (3, t)$, $t \geq 0$, is feasible and $x_1 + x_2 = 3 + t \rightarrow \infty$. There is no finite optimum. The best BFS is $(3,0)$ (in equational form $A = [1 \ 0 \ 1]$, the only bases are $\{x_1\}$ and $\{x_3\}$, since the x_2 column is zero), yet feasible points achieve arbitrarily larger values: the theorem's conclusion fails because its hypothesis fails. Same region, different objective: $\max x_1$ is bounded by 3.

Gotcha

- An *unbounded feasible region* does not imply an *unbounded objective*. The region above is unbounded for both objectives; only one objective is unbounded.
- “Best BFS” is meaningful only when the theorem applies. Without boundedness, the best BFS can be beaten by non-basic points.

Summary: Enumeration

- Under feasibility and boundedness, comparing all BFSs is a correct algorithm.
- It costs up to $\binom{n}{m}$ linear solves: fine for $\binom{4}{2}$, hopeless for $\binom{40}{20}$.
- Check boundedness with a certificate (a nonnegative combination bounding the objective) or refute it with a feasible improving ray (Proposition 3.2.4).

3.3 The Simplex Method

Motivation

Can we reach an optimum without examining every column selection? The simplex method (Dantzig, 1947) answers yes: start at a BFS, use the objective to pick a direction in which the objective increases, move along it until a nonnegativity constraint would be violated, and thereby arrive at a neighbouring BFS with a better value. Repeat until the algebra certifies optimality. Each step updates one basis, exchanging a single variable; no list of all bases is ever formed.

Instead of visiting every corner, walk from corner to corner, always uphill, and let the algebra tell you when no uphill direction remains.

A worked run

Example 3.3.1 (Setting up).

$$\max x_1 + x_2 \quad \text{s.t.} \quad -x_1 + x_2 \leq 1, \quad x_1 \leq 3, \quad x_2 \leq 2, \quad x_1, x_2 \geq 0.$$

The origin is feasible and $x_1 + x_2 \leq 3 + 2 = 5$ bounds the objective, so an optimal BFS exists. Slacks $x_3, x_4, x_5 \geq 0$ give

$$-x_1 + x_2 + x_3 = 1, \quad x_1 + x_4 = 3, \quad x_2 + x_5 = 2,$$

with $x_3 = 1 + x_1 - x_2$, $x_4 = 3 - x_1$, $x_5 = 2 - x_2$: each slack measures the gap between the two sides of its inequality.

Definition 3.3.2 (Tableau). A *tableau* (also called a *dictionary*) for a basis B is the system $Az = b$ rewritten so that each basic variable is expressed in terms of the nonbasic variables only, together with the objective expressed in the nonbasic variables:

$$x_i = p_i + \sum_{j \in N} q_{ij} x_j \quad (i \in B), \quad v = v_0 + \sum_{j \in N} r_j x_j.$$

The tableau is *feasible* if $p_i \geq 0$ for all $i \in B$. Setting every nonbasic variable to zero reads off the current BFS: $x_j = 0$ ($j \in N$), $x_i = p_i$ ($i \in B$), $v = v_0$. The numbers r_j are the *reduced costs*.

Example 3.3.3 (Tableau 0). Basic x_3, x_4, x_5 ; nonbasic x_1, x_2 :

$$\begin{aligned}x_3 &= 1 + x_1 - x_2 \\x_4 &= 3 - x_1 \\x_5 &= 2 \quad - x_2 \\v &= 0 + x_1 + x_2\end{aligned}$$

BFS $x = (0, 0, 1, 3, 2)$, $v = 0$. Both reduced costs are $+1$. Choose x_2 to enter (either choice is allowed); keep $x_1 = 0$. The rows give $x_3 = 1 - x_2 \geq 0$ and $x_5 = 2 - x_2 \geq 0$, and x_4 does not depend on x_2 . So x_2 can increase to 1, at which point x_3 reaches zero first: x_2 enters, x_3 leaves.

Example 3.3.4 (Pivot 1). Solve the leaving row for the entering variable: $x_3 = 1 + x_1 - x_2 \iff x_2 = 1 + x_1 - x_3$. Substitute into every other row, including the objective:

$$\begin{aligned}x_2 &= 1 + x_1 - x_3 \\x_4 &= 3 - x_1 \\x_5 &= 2 - (1 + x_1 - x_3) = 1 - x_1 + x_3 \\v &= x_1 + (1 + x_1 - x_3) = 1 + 2x_1 - x_3\end{aligned}$$

New nonbasic variables x_1, x_3 ; set them to zero: $x = (0, 1, 0, 3, 1)$, $v = 1$.

Example 3.3.5 (Pivot 2). Only x_1 has a positive reduced cost ($+2$). Keep $x_3 = 0$. Limits: $x_4 = 3 - x_1$ allows $x_1 \leq 3$; $x_5 = 1 - x_1$ allows only $x_1 \leq 1$; $x_2 = 1 + x_1$ does not limit. x_1 enters, x_5 leaves. From $x_5 = 1 - x_1 + x_3$: $x_1 = 1 + x_3 - x_5$.

$$\begin{aligned}x_1 &= 1 + x_3 - x_5 \\x_2 &= 1 + (1 + x_3 - x_5) - x_3 = 2 - x_5 \\x_4 &= 3 - (1 + x_3 - x_5) = 2 - x_3 + x_5 \\v &= 1 + 2(1 + x_3 - x_5) - x_3 = 3 + x_3 - 2x_5\end{aligned}$$

BFS $x = (1, 2, 0, 2, 0)$, $v = 3$.

Example 3.3.6 (Pivot 3: even a slack can enter). The reduced cost of x_3 is $+1$. Keep $x_5 = 0$. Only $x_4 = 2 - x_3$ limits: $x_3 \leq 2$; meanwhile $x_1 = 1 + x_3$ increases and $x_2 = 2$ is fixed. x_3 enters, x_4 leaves. From $x_4 = 2 - x_3 + x_5$: $x_3 = 2 - x_4 + x_5$.

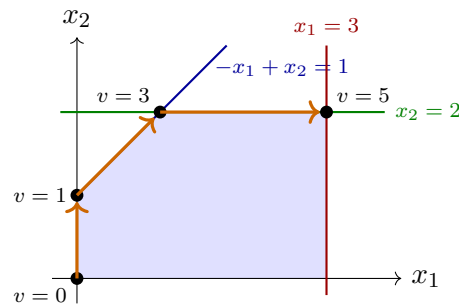
$$\begin{aligned}x_1 &= 1 + (2 - x_4 + x_5) - x_5 = 3 - x_4 \\x_2 &= 2 - x_5 \\x_3 &= 2 - x_4 + x_5 \\v &= 3 + (2 - x_4 + x_5) - 2x_5 = 5 - x_4 - x_5\end{aligned}$$

BFS $x = (3, 2, 2, 0, 0)$, $v = 5$.

Proposition 3.3.7 (Why we can stop: the objective row is a certificate). *In the final tableau, $v = 5 - x_4 - x_5$ holds for every feasible solution, and $x_4, x_5 \geq 0$, so $v \leq 5$ for all feasible points. The BFS $(x_1, x_2) = (3, 2)$ attains 5; hence it is optimal. Moreover $v = 5$ forces $x_4 = x_5 = 0$, which fixes every other variable through the tableau rows, so the optimum is unique.*

Proof. Each pivot rewrote the equations $Az = b$ by solving one equation for one variable and substituting; these are invertible operations, so the final system has exactly the same solution set as the original (Proposition 2.3.2). The objective row is an identity valid on that solution set. The rest is reading off signs. $\square$

Geometrically the run visited $(0, 0) \rightarrow (0, 1) \rightarrow (1, 2) \rightarrow (3, 2)$: four corners of the polygon, each adjacent to the previous one, with the objective climbing $0, 1, 3, 5$.



The general step

Consider a feasible tableau as in Definition 3.3.2.

Theorem 3.3.8 (Optimality test). *If every reduced cost satisfies $r_j \leq 0$, the current BFS is optimal. If moreover $r_j < 0$ for all $j \in N$, it is the unique optimal solution.*

Proof. For every feasible x , the objective row holds and $x_j \geq 0$, so $v = v_0 + \sum_{j \in N} r_j x_j \leq v_0$. The current BFS has value v_0 , hence it is optimal. If all $r_j < 0$, then $v = v_0$ forces $x_j = 0$ for every $j \in N$, and the basic rows then give $x_i = p_i$: the only feasible point with value v_0 is the current BFS. $\square$

Theorem 3.3.9 (Ratio test and unboundedness). *Suppose $r_j > 0$ for some $j \in N$, and increase $x_j = \theta \geq 0$ while keeping the other nonbasic variables at zero. Then each basic variable becomes $x_i = p_i + q_{ij}\theta$.*

1. *If $q_{ij} \geq 0$ for all $i \in B$, the LP is unbounded: the ray $\theta \mapsto x(\theta)$ is feasible for all $\theta \geq 0$ and $v = v_0 + r_j\theta \rightarrow \infty$.*
2. *Otherwise the largest feasible step is*

$$\theta^* = \min_{i \in B: q_{ij} < 0} \frac{p_i}{-q_{ij}} \geq 0,$$

and any row ℓ attaining the minimum has $x_\ell(\theta^) = 0$. The point $x(\theta^*)$ is feasible with objective $v_0 + r_j\theta^* \geq v_0$.*

Proof. The equations of the tableau hold along the whole line $x(\theta)$ by construction, so feasibility reduces to nonnegativity. Rows with $q_{ij} \geq 0$ satisfy $x_i(\theta) \geq p_i \geq 0$ for every $\theta \geq 0$; rows with $q_{ij} < 0$ require $\theta \leq p_i/(-q_{ij})$. If no row is of the second kind, all $\theta \geq 0$ are feasible and Proposition 3.2.4 applies with $c^\top d = r_j > 0$. Otherwise the binding constraint is the smallest ratio, which is ≥ 0 because $p_i \geq 0$; at θ^* the minimising row reaches exactly zero and no row is negative. $\square$

Key insight

Each basic row is a one-variable inequality $p_i + q_{ij}\theta \geq 0$. The ratio test is nothing more than solving all of them and taking the tightest. Zero ratios count: if some $p_i = 0$ with $q_{ij} < 0$, the step is $\theta^* = 0$.

Proposition 3.3.10 (Pivoting produces a new feasible tableau). *Let x_j enter and let x_ℓ be a row attaining θ^* (so $q_{\ell j} < 0$). Solving the row of x_ℓ for x_j and substituting into the other rows and the objective yields a tableau for the basis $B' = (B \setminus \{\ell\}) \cup \{j\}$, $N' = (N \setminus \{j\}) \cup \{\ell\}$. It is feasible, and its BFS is $x(\theta^*)$ with objective $v_0 + r_j\theta^*$.*

Proof. The row $x_\ell = p_\ell + q_{\ell j}x_j + \sum_{k \neq j} q_{\ell k}x_k$ can be solved for x_j because $q_{\ell j} \neq 0$: $x_j = -p_\ell/q_{\ell j} + x_\ell/q_{\ell j} - \sum_{k \neq j} (q_{\ell k}/q_{\ell j})x_k$. Substituting eliminates x_j from every other row, so each variable of B' is expressed in the variables of N' only, and the same equations still hold (Proposition 2.3.2). Setting N' to zero gives $x_j = -p_\ell/q_{\ell j} = p_\ell/(-q_{\ell j}) = \theta^*$ and, for $i \in B \setminus \{\ell\}$, $x_i = p_i + q_{ij}\theta^* \geq 0$ by the choice of θ^* ; these are exactly the coordinates of $x(\theta^*)$, hence nonnegative. The objective value at that point is $v_0 + r_j\theta^*$. $\square$

Technique: The simplex algorithm

1. Start with a feasible tableau (with all $b_i \geq 0$ in canonical form, the slack basis is one).
2. Read the objective row. If every reduced cost is ≤ 0 , stop: the current BFS is optimal (Theorem 3.3.8).
3. Choose an entering variable with $r_j > 0$.
4. Ratio test (Theorem 3.3.9). If no row limits the increase, stop: unbounded, return the ray. Otherwise the minimising row's variable leaves.
5. Pivot (Proposition 3.3.10) and return to step 2.

The LP never changes; only its description does.

Theorem 3.3.11 (Termination without degeneracy). *If every tableau visited is nondegenerate (all $p_i > 0$), then $\theta^* > 0$ at every pivot, the objective strictly increases, no basis is visited twice, and the algorithm terminates after finitely many pivots with an optimal BFS or a proof of unboundedness.*

Proof. With all $p_i > 0$ every ratio $p_i/(-q_{ij})$ is positive, so $\theta^* > 0$ and the objective increases by $r_j\theta^* > 0$. A basis determines its BFS and hence its objective value; a strictly increasing sequence of values cannot repeat a basis. There are at most $\binom{n}{m}$ bases, so the sequence is finite, and the only ways to stop are the two certificates. $\square$

Interpretation & discussion

Simplex is enumeration made lazy: it examines bases in an order chosen by the objective and typically visits a tiny fraction of them. Each visited basis costs a pivot, roughly the cost of one elimination step, rather than a full linear solve. The theorem also shows what *could* go wrong: degenerate pivots with $\theta^* = 0$ do not increase the objective, so the finiteness argument breaks. The next section treats the two exceptional cases.

Exceptional cases

Example 3.3.12 (Case 1: unboundedness). $\max x_1$ s.t. $x_1 - x_2 \leq 1$, $-x_1 + x_2 \leq 2$, $x \geq 0$. Slacks x_3, x_4 ; tableau 0:

$$x_3 = 1 - x_1 + x_2, \quad x_4 = 2 + x_1 - x_2, \quad v = x_1.$$

x_1 enters; only x_3 decreases, so $\theta^* = 1$ and x_3 leaves: $x_1 = 1 + x_2 - x_3$. After the pivot,

$$x_1 = 1 + x_2 - x_3, \quad x_4 = 2 + (1 + x_2 - x_3) - x_2 = 3 - x_3, \quad v = 1 + x_2 - x_3.$$

Now x_2 has reduced cost $+1$. Let $x_2 = \theta$ with $x_3 = 0$: $x_1 = 1 + \theta$ and $x_4 = 3$; no basic variable decreases. There is no leaving variable: the objective $v = 1 + \theta$ is unbounded. The certificate is the feasible ray $(x_1, x_2) = (1, 0) + \theta(1, 1)$, $\theta \geq 0$: both inequalities hold along it ($x_1 - x_2 = 1$, $-x_1 + x_2 = -1 \leq 2$).

Example 3.3.13 (Case 2: degeneracy, a zero step). $\max x_2$ s.t. $-x_1 + x_2 \leq 0$, $x_1 \leq 2$, $x \geq 0$. Tableau 0:

$$x_3 = x_1 - x_2, \quad x_4 = 2 - x_1, \quad v = x_2.$$

The initial BFS is degenerate: $x_3 = 0$. x_2 can enter, but with $x_1 = 0$ the row $x_3 = -x_2 \geq 0$ forces $\theta^* = 0$. Pivot anyway: $x_2 = x_1 - x_3$, giving

$$x_2 = x_1 - x_3, \quad x_4 = 2 - x_1, \quad v = x_1 - x_3.$$

The basis changed from $\{3, 4\}$ to $\{2, 4\}$, but the point $(0, 0, 0, 2)$ and the value 0 did not. A zero step does not prove optimality; it only re-describes the same vertex. Now x_1 has reduced cost $+1$: with $x_3 = 0$, $x_1 = \theta$, $x_2 = \theta$, $x_4 = 2 - \theta$, so $\theta^* = 2$ and x_4 leaves: $x_1 = 2 - x_4$,

$$x_1 = 2 - x_4, \quad x_2 = 2 - x_3 - x_4, \quad v = 2 - x_3 - x_4 \leq 2.$$

Optimum $(2, 2)$, $v^* = 2$. The first pivot changed the description; the second improved the solution.

Remark 3.3.14 (Cycling and how it is prevented). With degenerate pivots the objective can stall, and in rare, carefully constructed examples the algorithm can return to a basis it has already visited and loop forever. Simple entering/leaving rules prevent this. *Bland's rule*: among candidates, always pick the variable with the smallest index, both for entering and (among ties in the ratio test) for leaving. With Bland's rule the simplex method never cycles, so it always terminates. This is beyond the lecture; in exam-sized examples cycling does not occur, but zero

steps do, and you should recognise them.

A production decision: three juice blends

Example 3.3.15 (Model). A cooperative produces apple, berry and tropical juice. Per 100 litres: blending 1, 1, 1 hours (available 12); heat treatment 1, 2, 1 hours (available 18); imported concentrate 0, 1, 1 units of 10 L (available 8); profit 200, 400, 300 MAD. With x_1, x_2, x_3 in units of 100 L and v in hundreds of MAD,

$$\begin{aligned} \max \quad & v = 2x_1 + 4x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq 12, \quad x_1 + 2x_2 + x_3 \leq 18, \quad x_2 + x_3 \leq 8, \quad x \geq 0. \end{aligned}$$

The zero plan is feasible; $v \leq 4(x_1 + x_2 + x_3) \leq 48$ bounds the objective; an optimal BFS exists. Enumeration would test $\binom{6}{3} = 20$ selections. Slacks x_4, x_5, x_6 give the starting BFS $x = (0, 0, 0, 12, 18, 8)$, $v = 0$.

Example 3.3.16 (Three pivots). *Tableau 0.*

$$\begin{aligned} x_4 &= 12 - x_1 - x_2 - x_3 \\ x_5 &= 18 - x_1 - 2x_2 - x_3 \\ x_6 &= 8 - x_2 - x_3 \\ v &= 2x_1 + 4x_2 + 3x_3 \end{aligned}$$

Choose x_2 (largest reduced cost). With $x_1 = x_3 = 0$: $x_2 \leq 12, 9, 8$; x_6 leaves at 8. $x_2 = 8 - x_3 - x_6$.

Tableau 1.

$$\begin{aligned} x_2 &= 8 - x_3 - x_6 \\ x_4 &= 12 - x_1 - (8 - x_3 - x_6) - x_3 = 4 - x_1 + x_6 \\ x_5 &= 18 - x_1 - 2(8 - x_3 - x_6) - x_3 = 2 - x_1 + x_3 + 2x_6 \\ v &= 2x_1 + 4(8 - x_3 - x_6) + 3x_3 = 32 + 2x_1 - x_3 - 4x_6 \end{aligned}$$

BFS $(0, 8, 0, 4, 2, 0)$, $v = 32$. Only x_1 has a positive reduced cost. With $x_3 = x_6 = 0$: $x_4 = 4 - x_1$, $x_5 = 2 - x_1$; x_5 leaves at 2. $x_1 = 2 + x_3 - x_5 + 2x_6$.

Tableau 2.

$$\begin{aligned} x_1 &= 2 + x_3 - x_5 + 2x_6 \\ x_2 &= 8 - x_3 - x_6 \\ x_4 &= 4 - (2 + x_3 - x_5 + 2x_6) + x_6 = 2 - x_3 + x_5 - x_6 \\ v &= 32 + 2(2 + x_3 - x_5 + 2x_6) - x_3 - 4x_6 = 36 + x_3 - 2x_5 \end{aligned}$$

BFS $(2, 8, 0, 2, 0, 0)$, $v = 36$. x_3 enters. With $x_5 = x_6 = 0$: $x_1 = 2 + x_3$ increases, $x_2 = 8 - x_3$ allows 8, $x_4 = 2 - x_3$ allows only 2; x_4 leaves. $x_3 = 2 - x_4 + x_5 - x_6$.

Tableau 3.

$$\begin{aligned} x_1 &= 2 + (2 - x_4 + x_5 - x_6) - x_5 + 2x_6 = 4 - x_4 + x_6 \\ x_2 &= 8 - (2 - x_4 + x_5 - x_6) - x_6 = 6 + x_4 - x_5 \\ x_3 &= 2 - x_4 + x_5 - x_6 \\ v &= 36 + (2 - x_4 + x_5 - x_6) - 2x_5 = 38 - x_4 - x_5 - x_6 \end{aligned}$$

BFS $(4, 6, 2, 0, 0, 0)$, $v = 38$. All reduced costs are negative: optimal and unique (Theorem 3.3.8).

Example 3.3.17 (Interpretation). Produce 400 L apple, 600 L berry, 200 L tropical, for a daily profit of $100 \times 38 = 3,800$ MAD. All three slacks are zero: the plan uses all 12 blending hours, all 18 heat-treatment hours and all 80 L of concentrate. Check: $4 + 6 + 2 = 12$, $4 + 12 + 2 = 18$, $6 + 2 = 8$. The final objective row $v = 38 - x_4 - x_5 - x_6$ is also a certificate anyone can verify without redoing the pivots: it is the identity $2x_1 + 4x_2 + 3x_3 = 38 - (12 - x_1 - x_2 - x_3) - (18 - x_1 - 2x_2 - x_3) - (8 - x_2 - x_3)$, i.e. the weak-duality multipliers $y = (1, 1, 1)$ of Lemma 1.3.10: $A^\top y = (2, 4, 3) = c$ and $b^\top y = 38$.

Key insight

The negatives of the final reduced costs of the slack variables are exactly the weak-duality multipliers y . Simplex does not only find the optimal plan; it finds the certificate that proves it optimal. That is the seed of LP duality, developed later in the course.

Technique: Running simplex by hand without errors

- Keep every row in the form *basic* = *constant* + (*nonbasic terms*). A row with a basic variable on the right is a sign you forgot to substitute.
- *Ratio test*: only rows where the entering variable has a *negative* coefficient limit the step. Write the ratio next to each such row; include zero ratios.
- After a pivot, set the new nonbasic variables to zero and check every basic value is ≥ 0 . If not, you chose the wrong leaving row.
- Check the objective: the new v_0 must equal old $v_0 + r_j\theta^*$. In the juice example: $0 \rightarrow 32$ ($4 \cdot 8$), $32 \rightarrow 36$ ($2 \cdot 2$), $36 \rightarrow 38$ ($1 \cdot 2$).
- Verify the final answer against the *original* inequalities and read the slacks as unused resources.
- With several positive reduced costs, any choice is valid. The largest coefficient (Dantzig's rule) is the usual default; in exams, state your choice.

Classic problems

- The objective row reads $v = 10 + 3x_2 - x_4$ and the column of x_2 has no negative entry. Conclusion? Unbounded; the ray increasing x_2 is feasible with $v \rightarrow \infty$.
- The objective row reads $v = 10 - 3x_2 - x_4$. Conclusion? Optimal and unique, $v^* = 10$.
- The objective row reads $v = 10 + 0 \cdot x_2 - x_4$. Conclusion? Optimal, $v^* = 10$, but possibly not unique: increasing x_2 (if some row limits it) reaches another optimal solution, and the whole segment is optimal.

- Rows $x_3 = 0 - 2x_1 + x_2$ and $x_4 = 5 - x_1$; x_1 enters. Ratios $0/2 = 0$ and $5/1 = 5$; $\theta^* = 0$, x_3 leaves; a degenerate pivot.
- How many pivots can simplex take? At most the number of bases, $\binom{n}{m}$, with an anti-cycling rule; in practice far fewer.

Gotcha

- A positive coefficient of the entering variable in a basic row is *not* a limit: that variable increases. Only negative coefficients limit.
- Forgetting to substitute into the objective row leaves a stale certificate: the reduced costs must be recomputed at every pivot.
- “ $\theta^* = 0$ ” does not mean “optimal.” Check the reduced costs, not the step size.
- Simplex needs a *feasible* starting tableau. When some $b_i < 0$ the slack basis is infeasible and a preliminary phase is needed (not covered in this lecture).

Summary: The simplex method

- A tableau writes basic variables and the objective in terms of nonbasic variables; setting nonbasics to zero reads the BFS.
- All reduced costs ≤ 0 : optimal (strictly negative: unique). A positive reduced cost with no negative column entry: unbounded, with a ray certificate.
- Otherwise the ratio test picks the leaving variable and a pivot moves to an adjacent BFS with objective $\geq$ before; $>$ unless degenerate.
- Without degeneracy the method terminates; with it, an anti-cycling rule guarantees termination.
- The final reduced costs of the slacks are the weak-duality multipliers: simplex delivers an optimal plan and its proof.

Notes on the original slides

Every pivot in the Lecture 3 slides was recomputed (all four tableaux of the small example, the three of the juice example, the unbounded and degenerate examples), together with both enumeration tables and the binomial counts. All are correct. Points worth a reader’s attention:

- **Redundancy rule (slide 4).** The rule requires $\sum_i \lambda_i a_i = a$ exactly. When variables are nonnegative, a combination whose coefficients *dominate* a componentwise also works, because the bounds $-x_j \leq 0$ can be added with their own multipliers; the slide’s remark “you may also use variable bounds” covers this.
- **Tableau versus dictionary (slides 27–33).** What the slides call a tableau is the *dictionary* form (basic variables written in terms of nonbasic ones). Many textbooks reserve “tableau” for a matrix layout with the same information; the arithmetic is identical.

- **“The slack columns form I_3 : this is a BFS” (slide 27).** The identity columns make the slack set a *basis*; it is a *feasible* basis because every right-hand side is nonnegative. When some $b_i < 0$, the slack basis is not feasible and simplex cannot start from it.
- **Choice of entering variable (slide 28).** Both reduced costs equal 1; the slides choose x_2 . Choosing x_1 is equally valid and gives a different path to the same optimum. The theorems require only $r_j > 0$.
- **Termination (slide 40).** The slides state the algorithm without discussing whether it must stop. Theorem 3.3.11 proves termination when no pivot is degenerate; the remark after Example 3.3.13 explains why degenerate pivots need an anti-cycling rule.
- **Uniqueness (slide 34).** The argument “ $v = 5$ forces $x_4 = x_5 = 0$ ” is correct and general: strictly negative reduced costs imply a unique optimum (Theorem 3.3.8). A zero reduced cost does not.
- **Slide 21.** The best BFS of $\max x_1 + x_2$ s.t. $x_1 \leq 3$ is $(3, 0)$ because the x_2 column in equational form is the zero vector, so no basis contains x_2 .

Chapter takeaways

Summary: Solving linear programs

- **Prepare:** define decisions, units, constraints and objective; optionally prove and remove redundant inequalities; add one slack per inequality.
- **Enumerate** only for tiny problems: a feasible, bounded LP has an optimal BFS, but $\binom{n}{m}$ grows explosively.
- **Simplex:** start from a feasible tableau; enter a variable with positive reduced cost; leave by the ratio test; pivot; stop when the objective row certifies optimality or reveals an improving ray.
- **Exceptions:** no limiting row means unbounded; a zero step means degeneracy, which changes the basis but not the point.
- **Read the answer** in the original units, and read the final objective row as a proof.